Unit: 6 Unit 7: Factoring
Topic: Finding Common Factors

Name: ____________
Date: ____________

Advanced (Habanero)

Q1. What is the greatest common factor of 12 and 18?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12

Q2. Factor out the greatest common factor from $8x + 12y$
(A) $2(4x + 6y)$
(B) $4(2x + 3y)$
(C) $8(x + y)$
(D) $12(x + y)$
(E) $2(4x + 6y)$

Q3. What is the factored form of $9x^2 + 12x$
(A) $3x(3x + 4)$
(B) $3x(3x - 4)$
(C) $3x(3x + 4x)$
(D) $9x(x + 4)$
(E) $12x(x + 3)$

Q4. Factor $24a^2b + 30ab^2$
(A) $6ab(4a + 5b)$
(B) $6ab(4b + 5a)$
(C) $12ab(2a + 5b)$
(D) $6a(4ab + 5b)$
(E) $2ab(12a + 15b)$

Q5. Simplify the expression $4x^2y + 12xy^2$ by factoring out the greatest common factor.
(A) $4xy(x + 3y)$
(B) $4xy(x - 3y)$
(C) $12xy(x + y)$
(D) $4x(y^2 + 3y)$
(E) $2xy(2x + 6y)$

Q6. Factor out the greatest common factor from $15x^3y + 25x^2y^2$
(A) $5xy(3x^2 + 5y)$
(B) $5xy(3x^2 - 5y)$
(C) $5x(3x^2y + 5y^2)$
(D) $5xy(5x^2 - 3y)$
(E) $5x^2y(3x + 5y)$

Q7. Simplify $18x^4 + 12x^3$ by factoring out the greatest common factor.
(A) $6x^3(3x + 2)$
(B) $6x^3(3x - 2)$
(C) $12x^2(3x + 1)$
(D) $6x(3x^3 + 2x^2)$
(E) $6x^3(2x + 3)$

Q8. Factor the expression $10a^2b^2 + 15a^2b$
(A) $5a^2b(2b + 3)$
(B) $5a^2(2b^2 + 3b)$
(C) $5ab(2ab + 3a)$
(D) $10ab(ab + 3)$
(E) $5a^2b(3b - 2)$

Open Ended Questions (FRQ)

Q9. Factor $8x^2y + 12xy^2 + 16x^2$ and simplify.

Q10. Factor out the greatest common factor of $12x^3y + 9x^2y^2 + 6x^2y$ and write the simplified form.

Chef's Tips

- Tip 1: Emphasize the importance of factoring out the greatest common factor
- Tip 2: Review the distributive property to help students simplify expressions
- Tip 3: Provide examples of word problems to make factoring more applicable